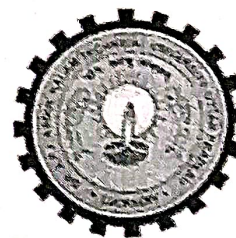




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A

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Deep Learning**

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NOTATIONS / NOMENCLATURE

Term	Description
SMILES	Simplified Molecular Input Line Entry System – text format to encode molecular structures
VAE	Variational Autoencoder – generative neural network
Latent Space	Compressed feature space learned by model
RDKit	Cheminformatics toolkit used for molecule validation
LogP	Partition coefficient – measure of lipophilicity
MW	Molecular Weight
HBD/ HBA	Hydrogen Bond Donor/Acceptor

ABSTRACT

The discovery of novel drug candidates is a complex, time-consuming, and expensive process due to the enormous size of chemical space and the limited feasibility of experimentally screening large numbers of molecular structures. Traditional computational and experimental approaches can explore only a small portion of the available chemical space. This project presents an AI-driven framework for de novo molecular generation using Deep Learning and Generative Artificial Intelligence to computationally explore chemical space and generate novel drug-like molecular structures.

The proposed framework employs an LSTM-based Variational Autoencoder (VAE) with a bidirectional LSTM encoder trained on the ZINC250K dataset containing 249,455 molecules represented as SMILES sequences. The model incorporates step-wise latent injection into the decoder, enabling direct access to the learned latent representation during sequence generation. An auxiliary molecular-property prediction component is used to encourage the latent space to retain chemically relevant information, while β -KL annealing is employed to balance reconstruction learning and latent-space regularization. During molecular generation, stochastic latent sampling with temperature scaling and Top-p (nucleus) sampling is used to generate diverse molecular structures.

The generated molecules are evaluated using RDKit for chemical validity and canonicalization, along with Lipinski's Rule of Five, QED, synthetic accessibility (SA), uniqueness, novelty, and internal diversity to assess their computational drug-like characteristics. From 10,000 generated SMILES sequences, 99.02% were chemically valid, 99.75% were unique, and 99.65% were novel with respect to the ZINC250K dataset, while 84.57% internal diversity was achieved. Furthermore, 99.98% of the chemically valid molecules satisfied Lipinski's Rule of Five, with a mean QED score of 0.7719 and mean SA score of 2.63. These results demonstrate the potential of the proposed deep generative framework for computational exploration of chemical space and generation of novel, diverse, and drug-like molecular structures.

1. INTRODUCTION

Artificial Intelligence (AI) and Deep Learning (DL) have emerged as promising approaches for de novo drug design. Traditional drug discovery is challenging because of the enormous size and complexity of chemical space, making exhaustive screening computationally and experimentally impractical. Generative AI enables computational exploration of this space by learning molecular patterns from existing data and generating new molecular structures.

Variational Autoencoders (VAEs) and recurrent neural networks such as Long Short-Term Memory (LSTM) models have demonstrated potential for molecular generation using SMILES representations [2]–[4]. However, recurrent VAE architectures can suffer from posterior collapse, where the decoder underutilizes the learned latent representation [5].

This project proposes an LSTM-based VAE with a bidirectional encoder for molecular generation using the ZINC250K dataset. The framework incorporates step-wise latent injection, auxiliary molecular-property prediction, and β -KL annealing to improve latent representation learning and generation. Generated molecules are evaluated for chemical validity, uniqueness, novelty, diversity, and computational drug-likeness.

1.1 Motivation

The research is motivated by the need for efficient computational methods to explore chemical space and generate novel molecular structures. The proposed approach addresses limitations of sequence-based VAEs by using a bidirectional encoder for richer molecular representations and step-wise latent injection to provide the decoder with continuous access to latent information. An auxiliary prediction component for LogP, QED, and SA encourages chemically relevant information to be retained in the latent space. β -KL annealing is used to balance reconstruction and latent-space regularization during training

1.2 Research Gap

Existing molecular generative models face challenges related to latent-space utilization, KL regularization, and balancing generation performance with model complexity. In recurrent VAE-based molecular generation, posterior collapse can reduce effective latent-space utilization. Furthermore, conventional decoder designs may provide limited access to latent information during sequence generation.

This research addresses these gaps through:

1. Improved latent representation using a bidirectional LSTM encoder.
2. Step-wise latent injection to maintain latent information throughout decoding.
3. Auxiliary property prediction to encourage chemically meaningful latent representations.
4. β -KL annealing to balance reconstruction and KL regularization.

1.3 Research Objectives

The main objectives of this research are:

1. To preprocess and tokenize molecular SMILES data from the ZINC250K dataset.
2. To develop an LSTM-based VAE with a bidirectional encoder for molecular representation learning.
3. To improve latent-space utilization using step-wise latent injection.
4. To incorporate auxiliary prediction of LogP, QED, and SA scores.
5. To use β -KL annealing for effective VAE training.
6. To generate molecular structures using stochastic latent sampling and temperature-controlled Top-p sampling.
7. To evaluate generated molecules using validity, uniqueness, novelty, internal diversity, Lipinski compliance, QED, and SA score.

2. LITERATURE REVIEW

Recent research has increasingly explored deep generative models for de novo molecular design as a means of efficiently navigating the large chemical space. Variational Autoencoders (VAEs) have been used to represent discrete molecular structures in a continuous latent space, enabling molecular generation and latent-space exploration [2]. Recurrent neural network-based approaches, particularly LSTM-based models, have also demonstrated the ability to learn SMILES syntax and generate novel molecular structures [3], [4].

Several generative architectures have subsequently been developed to improve molecular generation and property control. Conditional VAEs enable molecular properties to be incorporated into the latent representation for controlled generation, while conditional β -VAE approaches focus on latent-space disentanglement and property optimization [6], [7]. Other approaches such as JT-VAE, MolGAN, MoFlow, and MolHF explore molecular generation using chemically structured graphs, adversarial learning, normalizing flows, and hierarchical representations, respectively [8]–[11].

Despite these advances, sequence-based molecular VAEs continue to face challenges related to effective latent-space utilization and KL-divergence regularization. Posterior collapse can occur when recurrent decoders rely heavily on the reconstruction objective and fail to adequately utilize the latent representation. Re-balancing the VAE loss has therefore been investigated as a strategy for improving latent-space learning and molecular generation performance [5].

The existing literature also demonstrates the importance of evaluating generated molecules beyond simple sequence validity. Molecular generation studies commonly consider chemical validity, uniqueness, novelty, and molecular properties, while tools such as RDKit provide computational validation of generated structures. These evaluation principles provide a basis for assessing the quality and drug-like characteristics of generated molecules in the present work.

Based on these developments, the present research focuses on an LSTM-based VAE with a bidirectional encoder, incorporating step-wise latent injection, auxiliary molecular-property prediction, and β -KL annealing to improve latent representation learning and molecular generation. This provides a sequence-based approach that addresses key limitations identified in previous VAE-based molecular generation studies while maintaining a comparatively streamlined architecture.

3. DATASET AND PREPROCESSING

The quality of molecular data plays an important role in training deep generative models, as the model learns molecular patterns directly from the structures provided during training. In this work, the ZINC250K dataset was selected as the molecular training corpus. The dataset contains 249,455 molecular structures represented in SMILES format and provides a sufficiently large collection of drug-like molecules for sequence-based molecular generation.

3.1 Dataset Used

The ZINC250K dataset consists of 249,455 molecular structures and was used throughout the training and evaluation framework. The molecules are represented using SMILES (Simplified Molecular Input Line Entry System), which provides a compact textual representation of molecular structures and is suitable for processing with recurrent neural networks.

3.2 Data Cleaning and Validation

The raw molecular data was first examined to ensure that the SMILES representations were suitable for model training. RDKit was used to validate molecular structures and identify missing or invalid inputs. Molecules with invalid structures would be excluded during this stage to ensure that the model was trained only on chemically valid molecular representations.

The valid molecules were then converted into canonical SMILES representations. Canonicalization provides a standardized representation for each molecular structure, which helps maintain consistency during subsequent processing. Duplicate molecular structures were also removed after canonicalization.

Following preprocessing, all 249,455 molecules were retained, indicating that no molecules had to be removed from the final dataset after the applied validation and standardization steps. No missing or empty SMILES strings were identified.

3.3 SMILES Tokenization

Since the proposed model processes molecules as sequential SMILES representations, the canonical SMILES strings were converted into sequences of molecular tokens using a regex-based tokenizer. The tokenizer identifies relevant elements and structural symbols within the SMILES representation so that they can be processed as individual model inputs.

Four special tokens were introduced during tokenization:

- <PAD> – used to equalize the length of molecular sequences.
- <START> – indicates the beginning of a SMILES sequence.
- <END> – indicates the end of a sequence.
- <UNK> – represents an unknown or unrecognized token.

The resulting vocabulary contained 66 tokens. Because molecular SMILES sequences have different lengths, padding was applied so that sequences within a batch could be processed using a common sequence length.

3.4 Training and Validation Split

After preprocessing and tokenization, the dataset was divided into training and validation subsets. A 90:10 split was used, resulting in:

- Training set: 224,509 molecules
- Validation set: 24,946 molecules

The training set was used to learn molecular sequence patterns and latent representations, while the validation set was used to monitor model performance during training and assess its behavior on molecules not included in the training subset.

3.5 Auxiliary Molecular Properties

Along with the SMILES sequences, selected molecular properties were retained as auxiliary targets for the property-prediction component of the proposed model. These properties include LogP, QED, and Synthetic Accessibility (SA) score.

The auxiliary prediction component takes the learned latent representation and predicts these molecular properties. This additional objective encourages the latent representation to retain information related to chemically relevant molecular characteristics rather than being optimized solely for SMILES reconstruction.

3.6 Prepared Dataset

After preprocessing, the final dataset consisted of 249,455 validated and standardized molecular structures, represented as tokenized SMILES sequences with a vocabulary of 66 tokens. The data was divided into training and validation sets and supplemented with the required molecular property targets.

This prepared dataset was subsequently used for training the proposed LSTM-based Variational Autoencoder, including its bidirectional encoder, latent representation learning, decoder, and auxiliary molecular-property prediction component.

4. GENERATIVE MODELS

Generative models provide the foundation for computational molecular generation by learning patterns from existing molecular structures and generating new molecular sequences. In molecular design, Variational Autoencoders (VAEs) have been widely explored for learning continuous latent representations of molecular structures and generating new molecules from the learned distribution [2]. Recurrent neural networks, particularly LSTM-based models, are well suited for SMILES-based generation because they can learn the sequential structure and syntax of molecular representations [3], [4].

4.1 Variational Autoencoder

A Variational Autoencoder consists primarily of an encoder, latent space, and decoder. The encoder transforms an input molecular representation into a probabilistic latent representation, while the decoder reconstructs the molecular sequence from a sampled latent vector. This continuous latent space allows the model to learn meaningful molecular representations and generate new structures through stochastic sampling [2].

However, recurrent VAE architectures can experience posterior collapse, where the decoder learns to reconstruct the input while making insufficient use of the latent representation. This can reduce the effectiveness of latent-space learning and molecular generation [5].

4.2 Proposed LSTM-Based VAE

The proposed model is an LSTM-based Variational Autoencoder with a two-layer Bidirectional LSTM encoder. Each SMILES token is first converted into a 256-dimensional embedding. The bidirectional encoder processes the molecular sequence in both directions, and the resulting hidden representations are combined and projected into a 128-dimensional latent distribution, represented by its mean and log variance.

During training, a latent vector is sampled using the reparameterization trick. The sampled representation is used to initialize the decoder and is additionally provided to the decoder at every generation step. This step-wise latent injection is implemented by concatenating the latent vector with the token embedding, allowing the decoder to continuously access molecular information from the latent space during sequence generation.

4.3 Auxiliary Property Prediction

To encourage the latent space to capture chemically relevant information, the proposed architecture includes an auxiliary multilayer perceptron (MLP) that predicts molecular properties from the latent representation. The model predicts LogP, QED, and SA score alongside the SMILES reconstruction task.

The overall training objective therefore combines SMILES reconstruction loss, KL-divergence loss, and molecular-property prediction loss. This enables the latent representation to capture both structural and chemically relevant information.

4.4 β -KL Annealing

KL-divergence regularization is important for maintaining a meaningful probabilistic latent space, but applying strong KL regularization from the beginning of training can interfere with reconstruction learning. To address this, the proposed model uses β -KL annealing, in which the β coefficient is gradually increased during the initial training stage until reaching a maximum value of 0.05. It is then maintained at this value for the remaining training epochs.

4.5 Molecular Generation

After training, new molecular structures are generated by sampling latent vectors from a standard normal distribution and decoding them autoregressively from the <START> token. To control the randomness and diversity of generation, temperature scaling (0.8) and Top-p/nucleus sampling ($p = 0.9$) are applied during decoding. A total of 10,000 SMILES sequences were generated for evaluation.

The proposed generative framework therefore combines bidirectional sequence encoding, latent-space sampling, step-wise latent injection, auxiliary property learning, and β -KL annealing to improve the generation of chemically valid and diverse molecular structures.

5. DEEP LEARNING

Deep Learning (DL) provides the foundation for the proposed molecular generation framework by enabling neural networks to learn complex patterns directly from molecular representations. In molecular research, deep learning has been applied to representations such as SMILES sequences and molecular graphs, enabling models to learn structural characteristics and relationships within chemical data [2]–[4].

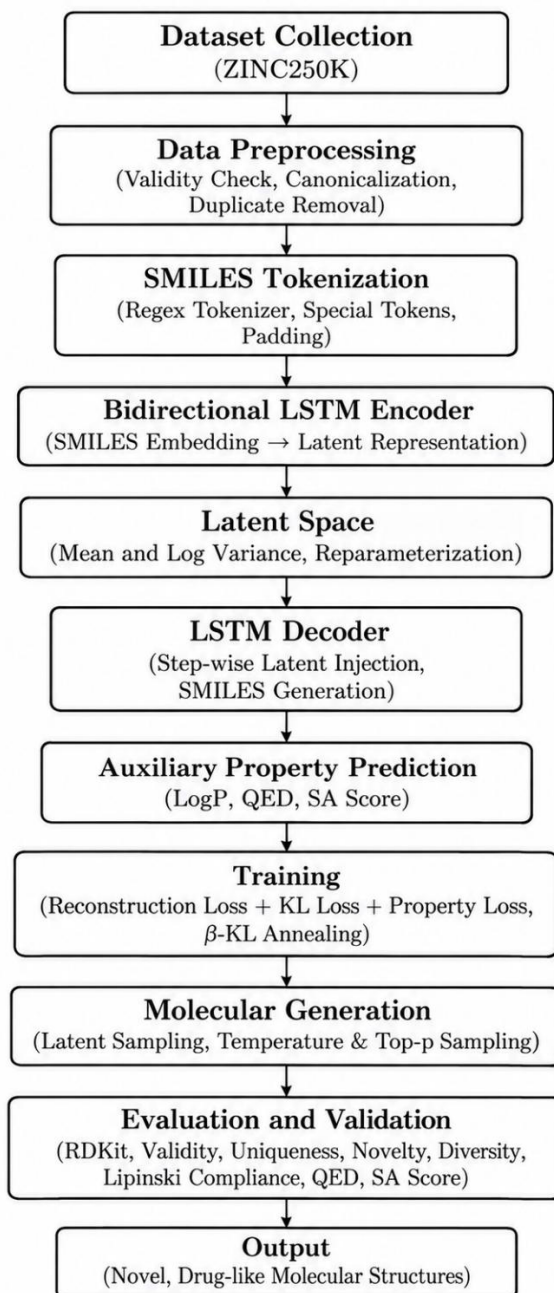
In this project, molecules are represented as SMILES sequences, which are processed using an embedding layer and a recurrent neural network. The proposed framework uses a two-layer Bidirectional LSTM to learn molecular representations from both forward and backward directions of the SMILES sequence. Each molecular token is converted into a 256-dimensional embedding, while the encoder uses a hidden dimension of 256. The resulting representations are combined to obtain the parameters of a 128-dimensional latent distribution.

The learned latent representation forms the core of the generative framework. It is used by the LSTM decoder for molecular sequence reconstruction and generation. Unlike a conventional encoder-decoder arrangement, the proposed model applies step-wise latent injection, providing the latent vector to the decoder along with the token embedding at each decoding step. This helps the decoder maintain access to the learned molecular representation throughout the generation process.

Deep learning is further utilized to capture chemically relevant information through an auxiliary molecular-property prediction network. The latent representation is used to predict LogP, QED, and SA score, encouraging the model to learn representations that contain meaningful molecular-property information in addition to sequence-level information.

Thus, deep learning in the proposed system serves as the basis for molecular representation learning, latent-space modeling, sequence reconstruction, property prediction, and molecular generation, with the LSTM-VAE architecture providing the primary generative mechanism.

6. METHODOLOGY



The proposed methodology develops an LSTM-based Variational Autoencoder (VAE) for de novo molecular generation using SMILES representations. The framework consists of molecular preprocessing, SMILES tokenization, latent representation learning using a bidirectional LSTM encoder, molecular reconstruction using an LSTM decoder, auxiliary molecular-property prediction, β -KL annealing during training, and stochastic molecular generation followed by computational evaluation.

6.1 Dataset Collection and Preparation

The ZINC250K dataset, containing 249,455 molecular structures, is used as the molecular training corpus. The molecules are represented in SMILES format. The dataset is processed using RDKit to verify chemical validity and identify missing or invalid molecular representations. Valid molecules are converted into canonical SMILES and duplicate structures are removed.

After preprocessing, the dataset is divided into 224,509 training molecules and 24,946 validation molecules, corresponding to a 90:10 split.

6.2 SMILES Tokenization

The standardized SMILES strings are converted into sequences using a regex-based tokenizer. Four special tokens, namely <PAD>, <START>, <END>, and <UNK>, are added to support sequence processing and molecular generation. The resulting vocabulary contains 66 tokens.

Since SMILES sequences have different lengths, padding is applied to obtain uniform-length sequences for batch processing. The resulting token sequences are then converted into embeddings before being passed to the neural network.

6.3 Bidirectional LSTM Encoder

The encoder forms the first major component of the proposed VAE. It consists of a two-layer Bidirectional LSTM that processes the molecular sequence in both forward and backward directions. Each input token is represented using a 256-dimensional embedding, and the encoder uses a hidden dimension of 256. The forward and backward hidden representations are combined and projected to obtain the parameters of a 128-dimensional latent distribution, represented by its mean and log variance.

This bidirectional processing enables the encoder to capture information from both directions of the molecular sequence, providing a richer representation for subsequent latent-space learning.

6.4 Latent Representation and Reparameterization

The encoder produces the mean and log variance of the latent distribution. During training, a latent vector is sampled using the reparameterization trick, which allows stochastic sampling while maintaining differentiability for backpropagation.

The resulting latent vector represents the learned molecular information and serves as the connection between the encoder and decoder. It is subsequently used to initialize the decoder and is also supplied directly during each decoding step.

6.5 LSTM Decoder with Step-wise Latent Injection

The sampled latent vector is used to initialize the hidden and cell states of a two-layer LSTM decoder. In addition, the proposed architecture incorporates step-wise latent injection, where the latent vector is concatenated with the token embedding at every decoding step.

This allows the decoder to maintain direct access to the latent molecular representation throughout sequence reconstruction and generation rather than relying only on the initial decoder state. This mechanism is intended to improve latent-space utilization during molecular sequence generation.

6.6 Auxiliary Molecular-Property Prediction

An auxiliary Multilayer Perceptron (MLP) is incorporated into the architecture to predict selected molecular properties from the latent representation. The model predicts LogP, QED, and SA score.

The auxiliary prediction task encourages the latent space to preserve chemically relevant molecular information while the decoder simultaneously learns to reconstruct the input SMILES sequence.

6.7 Training Objective and β -KL Annealing

The model is trained using a combined objective consisting of:

- SMILES reconstruction loss
- KL-divergence loss
- Molecular-property prediction loss

The reconstruction component measures how accurately the decoder reproduces the input SMILES sequence, while the KL-divergence term regularizes the learned latent distribution toward a standard normal distribution. The property-prediction loss measures the difference between predicted and target molecular properties.

To control the contribution of KL regularization during training, β -KL annealing is applied. The β coefficient is gradually increased during the initial training stage until reaching a maximum value of 0.05, after which it remains constant. The model is trained for 15 epochs using a learning rate of 0.001, batch size of 128, and the AdamW optimizer.

6.8 Molecular Generation

After training, new molecular structures are generated by sampling latent vectors from a standard normal distribution. The sampled vectors are passed to the decoder, which generates SMILES sequences autoregressively beginning with the <START> token.

To control the randomness and diversity of generated sequences, temperature scaling with a value of 0.8 and Top-p (nucleus) sampling with $p = 0.9$ are applied during decoding. A total of 10,000 SMILES sequences are generated in batches of 500 for subsequent evaluation.

6.9 Molecular Validation and Evaluation

The generated SMILES sequences are first evaluated using RDKit to determine their chemical validity. Canonical SMILES are then used to identify duplicate structures and calculate uniqueness. Exact-match novelty is determined by comparing the InChIKeys of generated molecules with those in the ZINC250K reference set. Molecular diversity is evaluated using Morgan fingerprints (ECFP4, radius 2, 2048 bits) and pairwise Tanimoto similarity. Internal diversity is calculated as one minus the mean pairwise similarity. To reduce computational cost, up to 2,000 unique molecules are randomly selected for this analysis.

Drug-like characteristics are evaluated using Lipinski's Rule of Five and QED, while SA score is used to assess synthetic accessibility.

7. EXPERIMENTATION SETUP AND RESULTS

The experimentation phase involved the implementation, training, and evaluation of the proposed LSTM-based Variational Autoencoder (VAE) for de novo molecular generation. The model was trained using the preprocessed ZINC250K dataset and evaluated based on molecular reconstruction and generation performance, including chemical validity, uniqueness, novelty, diversity, and computational drug-likeness.

7.1 Experimental Setup

The proposed framework was implemented in Python using PyTorch, with RDKit used for molecular validation, canonicalization, and molecular property calculations. The experiments were conducted in a GPU-enabled computational environment.

The model consists of a two-layer Bidirectional LSTM encoder, a 128-dimensional latent space, and a two-layer LSTM decoder. Each molecular token is represented using a 256-dimensional embedding with a hidden dimension of 256. The model was trained using the AdamW optimizer, with a learning rate of 0.001 and a batch size of 128 for 15 epochs. β -KL annealing was applied during training, with β increasing to a maximum value of 0.05.

7.2 Training Performance

The training and validation losses were monitored throughout the 15 training epochs. The training loss decreased from 0.9644 to 0.6604, while the validation loss decreased from 0.7097 to 0.6331. The reconstruction loss decreased from 0.9017 to 0.5899, indicating improved SMILES sequence reconstruction.

During training, the KL coefficient was gradually increased from 0.0067 at Epoch 1 to 0.05 at Epoch 8, after which it was maintained at 0.05. The KL-divergence loss decreased from 4.5323 to 0.7476 during training. The lowest validation loss of 0.6331 was obtained at Epoch 15.

7.3 Molecular Generation Results

After training, 10,000 SMILES sequences were generated using stochastic latent sampling and autoregressive decoding. Temperature scaling of 0.8 and Top-p sampling with $p = 0.9$ were used during generation.

The generated molecules demonstrated strong computational generation performance:

Evaluation Metric	Result
Molecules Generated	10,000
Chemical Validity	99.02%
Uniqueness	99.75%

Exact-Match Novelty	99.65%
Internal Diversity	84.57%
Lipinski Compliance*	99.98%
Mean QED	0.7719
Mean SA Score	2.63

*Calculated with respect to the 9,902 chemically valid molecules.

Out of the 10,000 generated sequences, 9,902 molecules were chemically valid, corresponding to a validity of 99.02%. Among these valid molecules, 9,877 were unique. Exact-match comparison with the ZINC250K reference set resulted in 99.65% novelty, while the internal diversity of the generated molecular set was 84.57%. Furthermore, 9,900 of the 9,902 valid molecules (99.98%) satisfied Lipinski's Rule of Five.

The generated molecules achieved a mean QED score of 0.7719, indicating favorable computational drug-likeness, while the mean SA score of 2.63 indicates the synthetic accessibility characteristics of the generated set.

7.4 Novelty Evaluation

The generated molecules achieved novelty rates of 99.65% against ZINC250K, 98.85% against ChEMBL 37, and 81.95% against PubChem. These results provide additional evidence that the model generated structures that were largely absent from the evaluated reference databases.

7.5 Comparison with Existing Models

The proposed model was also compared with representative molecular generative approaches reported in existing literature. The proposed model achieved 99.02% validity, 99.75% uniqueness, and 99.65% novelty, outperforming the reported validity and uniqueness values of several comparison models.

7.6 Discussion

The experimental results demonstrate that the proposed LSTM-based VAE can generate molecular structures with high chemical validity, uniqueness, and novelty while maintaining favorable computational drug-like properties. The combination of bidirectional encoding, step-wise latent injection, auxiliary property prediction, and β -KL annealing provides the framework with mechanisms for improved latent representation learning and molecular generation.

However, the obtained results represent computational evaluation only. They do not establish biological activity, therapeutic efficacy, or experimental synthesizability of the generated molecules. These aspects would require further laboratory and biological validation.

8. CONCLUSION

This project developed an LSTM-based Variational Autoencoder (VAE) with a Bidirectional LSTM encoder for de novo molecular generation using the ZINC250K dataset. The proposed framework combines bidirectional molecular representation learning, step-wise latent injection, auxiliary molecular-property prediction, and β -KL annealing to improve latent-space learning and molecular sequence generation.

The trained model generated 10,000 SMILES sequences, of which 99.02% were chemically valid, 99.75% were unique, and 99.65% were novel with respect to the ZINC250K dataset. The generated molecules achieved an internal diversity of 84.57%, while 99.98% of the chemically valid molecules satisfied Lipinski's Rule of Five. The generated set also obtained a mean QED score of 0.7719 and mean SA score of 2.63, indicating favorable computational drug-like characteristics.

These results demonstrate the potential of the proposed deep generative framework to computationally explore chemical space and generate novel, unique, diverse, and drug-like molecular structures. However, the results are based on computational evaluation and do not establish biological activity, therapeutic efficacy, or experimental synthesizability. Further work can therefore focus on property-conditioned molecular generation, toxicity assessment, biological evaluation, and experimental validation of generated candidates.

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